

Week 7 Day 4 Lab 4

Kafka Installation

Task 1 – Start Kafka

1. Download the latest version of Kafka from Apache Kafka
2. Check Java version `java -version` should be above Java 17+
3. Extract the Kafka folder using command
`tar -xzf kafka2.13-4.3.1.tgz` (hint:specify the folder where kafka is present user\ganas\Downloads\kafka2.13-4.3.1)
4. Go inside the extracted folder
`cd kafka_2.13-4.3.1`
5. Generate Cluster UUID
`bin\windows\kafka-storage.bat random-uuid` (This gives Kafka Cluster Id)
6. Format Storage Directory(KRaft Mode)
`bin\windows\kafka-storage.bat format --standalone -t $Kafka-cluster-id -c config\server.properties`
7. Start Kafka Server
`bin\windows\kafka-server-start.bat config\server.properties`(Leave that terminal as it is if it shows kafka started)

Task 2 – Inspect the Enviornment

Use the below command to to list the topics, if topics are already exist you get the list or else you will get the empty list

Open new Terminal

```
.bin\windows\kafka-topics.bat `
```

```
--bootstrap-server localhost:9092 `
```

```
--list
```

```
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
2026-08-20T12:22:57.047575400Z main ERROR Reconfiguration failed: No configuration found for '2c7b84de' at 'null' in 'null'
__consumer_offsets
ironhack
partition_example
```

Task 3 – Create the Topic

Open new Terminal

```
bin\windows\kafka-topics.bat --create --topic ironhack --partitions 1 --
replication-factor 1 --bootstrap-server localhost:9092
```

This will create a new topic called Ironhack

```
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-topics.bat --c
reate --topic ironhack --partitions 1 --replication-factor 1 --b
ootstrap-server localhost:9092
2026-08-20T10:17:48.157275700Z main ERROR Reconfiguration failed
: No configuration found for '2c7b84de' at 'null' in 'null'
Created topic ironhack.
```

Now if you do list you will see ironhack topic created

```
bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
```

```
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
2026-08-20T10:29:58.453404500Z main ERROR Reconfiguration failed
: No configuration found for '2c7b84de' at 'null' in 'null'
ironhack
```

Task 4 – Describe and interpret the topic

```
. \bin\windows\kafka-topics.bat `
```

```
--bootstrap-server localhost:9092 `
```

```
--describe `
```

```
--topic ironhack
```

```
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-topics.bat --describe --topic ironhack --bootstrap-server localhost:9092
2026-08-20T12:19:23.825571200Z main ERROR Reconfiguration failed: No configuration found for '2c7b84de' at 'null' in 'null'
Topic: ironhack TopicId: 9274QWAXRmab47mfAI2sfQ PartitionCount: 1 ReplicationFactor: 1 Configs: min.insync.replicas=1,segment.bytes=1073741824
    Topic: ironhack Partition: 0 Leader: 1 Replicas: 1 Isr: 1 Elr: LastKnownElr:
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-topics.bat --describe --topic partition_example --bootstrap-server localhost:9092
2026-08-20T12:19:49.630081900Z main ERROR Reconfiguration failed: No configuration found for '2c7b84de' at 'null' in 'null'
Topic: partition_example TopicId: twlHpA4ESNiF7sgJ2oo0zQ PartitionCount: 2 ReplicationFactor: 1 Configs: min.insync.replicas=1,segment.bytes=1073741824
    Topic: partition_example Partition: 0 Leader: 1 Replicas: 1 Isr: 1 Elr: LastKnownElr:
    Topic: partition_example Partition: 1 Leader: 1 Replicas: 1 Isr: 1 Elr: LastKnownElr:
```

Task 5 – Create a kafka console producer

```
bin\windows\kafka-console-producer.bat --topic ironhack --bootstrap-server localhost:9092
```

Add the data to see it in consumer

```
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-console-producer.bat --topic ironhack --bootstrap-server localhost:9092
2026-08-20T10:32:47.128427600Z main ERROR Reconfiguration failed
: No configuration found for '2c7b84de' at 'null' in 'null'
>this
>is
>a
>producer
>this
>is
>a
>message
>for
>consumer
>
```

Task 6 – Create the kafka console consumer

Open a new terminal

```
bin\windows\kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic ironhack --from-beginning
```

You can see the data inserted in kafka producer in kafka consumer and you can update the data in producer and see the data being reflected in consumer in live

```
C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic ironhack --from-beginning
2026-08-20T10:34:31.032345700Z main ERROR Reconfiguration failed
: No configuration found for '2c7b84de' at 'null' in 'null'
The consumer rebalance protocol (KIP-848) is production-ready! Set group.protocol=consumer to try it out. See https://kafka.apache.org/documentation/#consumer_rebalance_protocol
this
is
a
producer
this
is
a
message
for
consumer
```

<pre>--topic <String: topic> e to produce --version C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-console-producer.bat --topic ironhack --bootstrap-server localhost:9092 2026-08-20T10:32:47.128427600Z main ERROR Reconfiguration failed : No configuration found for '2c7b84de' at 'null' in 'null' >this >is >a >producer >this >is >a >message >for >consumer >this >data >produces >in >live >in >the >consumer >it >helps >in >live >streaming >of >data ></pre>	<pre>REQUIRED: The topic name messages to. Display Kafka version. C:\Users\ganas\kafka_2.13-4.3.1>bin\windows\kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic ironhack --from-beginning 2026-08-20T10:34:31.032345700Z main ERROR Reconfiguration failed : No configuration found for '2c7b84de' at 'null' in 'null' The consumer rebalance protocol (KIP-848) is production-ready! Set group.protocol=consumer to try it out. See https://kafka.apache.org/documentation/#consumer_rebalance_protocol this is a producer this is a message for consumer this data produces in live in the consumer it helps in live streaming of data ></pre>
--	---